

Vision

The application proposes ambitious research: the comprehensive development of Directed Type Theory (DTT) as a new foundation for (mechanised) mathematics and computer science.

For more than 10 years, Homotopy Type Theory (HoTT) has been generating interest in the foundations of mathematics within the research community. The proposal takes into account relevant recent developments and builds on the need (justified in recent work) to go beyond HoTT symmetry (isomorphisms), towards a more directed approach (morphisms).

STRENGTH. For the foundational aspects above, the applicant's justifications of excellence, potential, and timeliness are more than satisfactory. We deem the proposed research potentially groundbreaking.

STRENGTH. About the impact on the research fields of type theory, category theory, programming languages, and formalised mathematics, the proposal clearly explains how a comprehensive DTT development would unify existing attempts and shape future research directions in the field.

From a broader perspective, the applicant well delineates what areas will be impacted by their research. By developing a workable theory, through mechanisation, both research in the foundations of mathematics and hardware and software verification will see tangible benefits.

MINOR WEAKNESS. However, it is not clear to us how DTT specifically helps with respect to other foundational approaches. While the quantum-physics application from the Approach section provides strong motivation, we would have still liked a bit more detail about how DTT compares to other logical foundations, e.g., for the development of proof assistants.

QUESTIONS.

Could you please comment on some envisioned applications of DTT? For example, what specific concepts of mathematics will DTT help capture, unify, ...? Or, what specific application problem will DTT help solve (for example, in the design and implementation of proof assistants)?

Approach

STRENGTH. The state of the art is very clearly presented, as is the direction that the proposed research will take from this.

STRENGTH. The proposal is convincing in terms of methodology and effectiveness. The Approach description is well-structured in work packages. Every work package clearly delineates its aims and a strategy to achieve those. Particularly, the expected output strategy is made explicit and seems realistic in the amount and typology of declared deliverables.

The proposed research is foundational and, as such, it presents uncertainties and challenges. The applicant discusses risks and mitigations for each work package, making

explicit the possibility of partial results. We consider this, together with the aforementioned awareness of the state of the art and a clear description of the theoretical approaches they intend to adopt, a STRENGTH in terms of feasibility of the project. The application to quantum physics seems a high-risk, high-reward endeavour worth exploring.

However, about feasibility, it is also where we identify a WEAKNESS of this proposal: strong dependencies are described in the work plan page. As the applicant acknowledges, most work packages are interconnected. In particular, complete success for WP2 seems necessary to the development of WP3, WP4, WP6, and WP7. Given its central role, we would have liked a more detailed mitigation strategy for WP2 (see questions below).

QUESTIONS.

From WP2, Risk: “Even if the general case remains out of reach, we expect to develop robust semantic frameworks for key fragments, which will be sufficient for our intended applications and provide a solid foundation for future work.”

Could you expand on this? For WP2, 4 different directions to explore are listed; what are the implications of success in one or the other? How will these affect the following work packages?

Applicant and Team Capability to Deliver

STRENGTH. In short, the applicant excellently argues how they, their team, and collaborators have the full capability to deliver.

The applicant and their team have been at the top of this field since HoTT was born. Their experience, skills, and expertise are undeniable. They have also convincingly argued about their network of collaborators and the support they can offer.

The applicant demonstrates their (and their team's) past engagement in projects where their leadership and management were key, and their commitment to the development of others. These skills and engagement have often been transferred to the wider community and aimed at creating a positive research environment (Module 3 and Module 4 of the application).

Ethics and Responsible Research and Innovation (RRI)

The proposed research is foundational in the fields of mathematics and theoretical computer science. There are no major ethical issues connected to this kind of research.

STRENGTH. The applicant, however, shows awareness of the importance of RRI and the ethical responsibilities of any researcher. They clearly state this and describe the actions they will take to ensure that their research is ethical and responsible.

Resources and Cost Justification

As acknowledged by the applicant, the largest component of the requested resources is staff time, with some additional travel-and-subsistence costs to enable collaboration, dissemination, and community engagement.

This perfectly aligns with the foundational research that the applicant proposes: we find their request more than appropriate and their justifications convincing. As far as we can

tell, the resource and cost requests are optimised towards achieving the intended outcomes and maximising impact.

STRENGTH. In particular, the applicant proposes foundational and potentially groundbreaking theoretical research that is high-risk in some of its aspects. Directing resources towards people (staff and collaborations) seems only reasonable.

Score: 6